Briefing

Adaptive lasso; graph class, edge weights, and condition (W2)

Joon Kim

October 2026

Adaptive lasso: estimator and weights

Model [1]: $X_1, \dots, X_n$ i.i.d. $N(0, \Sigma^*)$, $M^*\Sigma^* + \Sigma^* M^{*\top} + C = 0$, M^* stable, C known

Direct loss [1], with $m = \mathrm{vec}(M)$, whose entries are indexed by j, k:

$$L(m) = \tfrac{1}{2}\left\| M\hat{\Sigma} + \hat{\Sigma}M^\top + C \right\|_F^2 = \tfrac{1}{2}\, m^\top \hat{\Gamma}\, m - \hat{g}^\top m + \text{const}, \qquad \hat{\Gamma} = \Gamma(\hat{\Sigma}), \quad \hat{g} = g(\hat{\Sigma})$$

Pilot $\hat{M}^0$: the lasso at the smallest λ of its grid, every entry penalised

Weights: $\hat{w} = \max |\hat{M}^0| \,/\, |\hat{M}^0|$ entrywise, the maximum over all entries; $\hat{w}_k = \infty$ (so $m_k = 0$) where $\hat{M}^0 = 0$

adaptive lasso [2], every entry penalised, the diagonal too:

$$\hat{M} \in \arg\min_m \; L(m) + \lambda \sum_k \hat{w}_k \, |m_k|$$

Notation in W.1–W.3

$S = \mathrm{supp}(m^*)$, the diagonal included; $d = |S|$; $\beta_{\min} = \min_{k \in S} |m^*_k|$; $\hat{S} = \mathrm{supp}(\hat{M})$

Weights on and off S:

$$\bar{w}_S = \max_{k \in S} \hat{w}_k, \qquad \underline{w} = \min_{j \in S^c} \hat{w}_j, \qquad r = \frac{\bar{w}_S}{\underline{w}} = \frac{\max_{j \in S^c} |\hat{M}^0_j|}{\min_{k \in S} |\hat{M}^0_k|}$$

Irrepresentability, with $\Gamma^* = \Gamma(\Sigma^*)$ and $P = \Gamma^*_{S^c S} \left(\Gamma^*_{SS}\right)^{-1}$:

$$\rho = |||P|||_\infty = \max_{j \in S^c} \sum_{k \in S} |P_{jk}| \quad \text{(the lasso's [1])}, \qquad \rho_w = \max_{j \in S^c} \frac{1}{\hat{w}_j} \sum_{k \in S} |P_{jk}| \, \hat{w}_k \quad \text{(weighted [3])}$$

Constants of [1]: $c_{\Gamma^*} = |||(\Gamma^*_{SS})^{-1}|||_\infty$, $c_{M^*} = \|m^*\|_\infty$; errors $\Delta_\Gamma = \hat{\Gamma} - \Gamma^*$, $\Delta_g = \hat{g} - g^*$, with $g^* = g(\Sigma^*)$

W.1: deterministic, after [1, Theorem D.1] - PDW

Let M^* be stable with support S, let the weights have $\bar{w}_S < \infty$, and fix any $\kappa \geq 1 + \rho$.
Assume:

(W1) Γ^*_{SS} is invertible

(W2) $\rho_w \leq 1 - \alpha$ for some $\alpha \in (0, 1]$

(W3) $\|\|(\Delta_\Gamma)_{\cdot S}\|\|_\infty \leq \varepsilon_1$ (the columns in S) and $\|\Delta_g\|_\infty \leq \varepsilon_2$, with $\varepsilon_1 \leq \alpha \,/\, \big(3c_{\Gamma^*}(\kappa r + \alpha)\big)$

(W4) $\lambda\, \underline{w} > (3\kappa/\alpha)\, \max\{c_{M^*}\varepsilon_1,\ \varepsilon_2\}$

Then:

(a) $\hat{M}$ is unique, $\hat{S} \subseteq S$, and
$$\|\hat{M} - M^*\|_\infty \leq b_w := \frac{c_{\Gamma^*}}{1 - c_{\Gamma^*}\varepsilon_1}\big(c_{M^*}\varepsilon_1 + \varepsilon_2 + \lambda \bar{w}_S\big) < c_{\Gamma^*}\lambda\Big(\bar{w}_S + \frac{\alpha}{\kappa}\,\underline{w}\Big)$$

(b) if $\beta_{\min} > b_w$, then $\hat{S} = S$, with the signs of M^*

Changed from [1, Theorem D.1]: the weights $\hat{w}$ in the penalty; ρ_w for ρ in (W2); κ for $2 - \alpha$; $\kappa r + \alpha$ for 2 in (W3); $\lambda\underline{w}$ for λ in (W4). With $\hat{w} \equiv 1$ and $\kappa = 2 - \alpha$, W.1 reduces to [1, Theorem D.1].

W.2: probabilistic, after [1, Theorem 2]

In the setting of [1, Theorem 2], let $\kappa \geq 1+\rho$, $\alpha \in (0,1]$ and $\bar{r} \geq 0$.
Let $\mathcal{E}_w$ be an event on which $\bar{w}_S < \infty$, $\rho_w \leq 1-\alpha$ and $r \leq \bar{r}$.

$$\varepsilon = \sqrt{\tau_1 \tilde{c}\, d\, p \,/\, n}, \qquad c_w = \frac{3 c_{\Gamma^*}(\kappa \bar{r} + \alpha)}{\alpha}$$

Assume $n > \tau_1 \tilde{c}\, d\, p \max\{c_w^2, 1/4\}$ for some $\tau_1 > 1$, and $\lambda \underline{w} > (3\kappa/\alpha)\max\{c_{M^*}, 1\}\,\varepsilon$ on $\mathcal{E}_w$.
Then, with probability at least $1 - c_2 e^{-\tau_1 p} - P(\mathcal{E}_w^c)$:

- (a) and (b) of W.1 holds

Changed from [1, Theorem 2]: c_w for $c_* = 6c_{\Gamma^*}/\alpha$; the probability loses $P(\mathcal{E}_w^c)$.
With $\hat{w} \equiv 1$, $\kappa = 2 - \alpha$ and $\bar{r} = 1$, W.2 reduces to [1, Theorem 2].

Corollary W.3: a pilot close to the truth

Let $\|\hat{M}^0 - M^*\|_\infty \le \delta < \beta_{\min}/(1+\rho)$, and put

$$\bar{r} = \frac{\delta}{\beta_{\min} - \delta}, \qquad \alpha = 1 - \rho\,\bar{r}.$$

Then $\alpha \in (0, 1]$, and:

(i) $\bar{w}_S < \infty$: all infinite weights lie in S^c

(ii) $r \le \bar{r}$

(iii) $\rho_w \le \rho\, r \le 1 - \alpha$, so (W2) holds

Hence:

- W.1 applies with this α whenever (W1), (W3) and (W4) hold;
- W.2 applies with $\mathcal{E}_w = \{\|\hat{M}^0 - M^*\|_\infty \le \delta\}$ and this $\bar{r}$.

Graph class

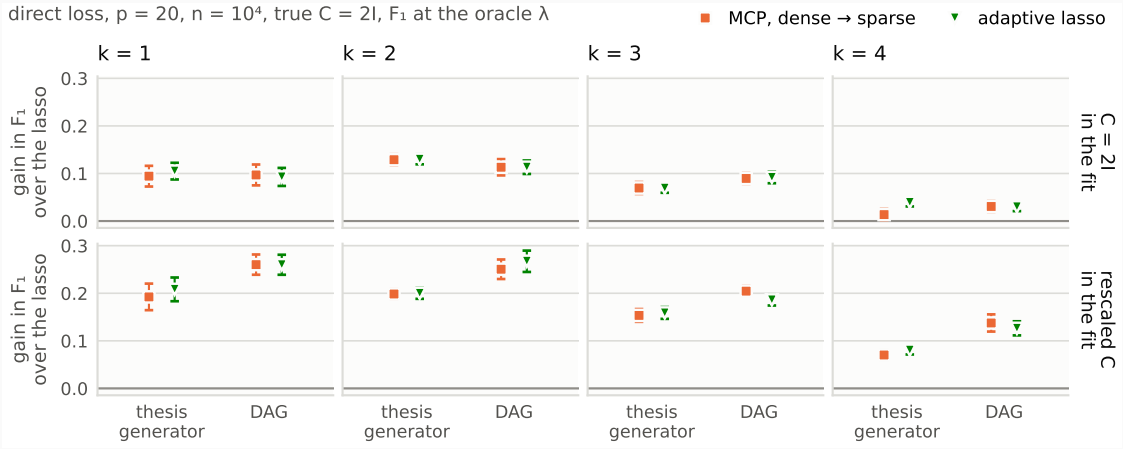
- thesis generator: $P(M_{ij} \neq 0) = k/p$ for every pair $i \neq j$
- DAG: a random order π of the nodes; $P(M_{ij} \neq 0) = 2k/p$ if $\pi(j) < \pi(i)$, else 0

Edge weights on the support

- thesis generator: $M_{ij} \sim N(0, 1)$
- bounded away from 0: $|M_{ij}| \sim U[w_{\min}, 1]$ with a random sign, $w_{\min} \in \{0, 0.1, 0.25, 0.5\}$

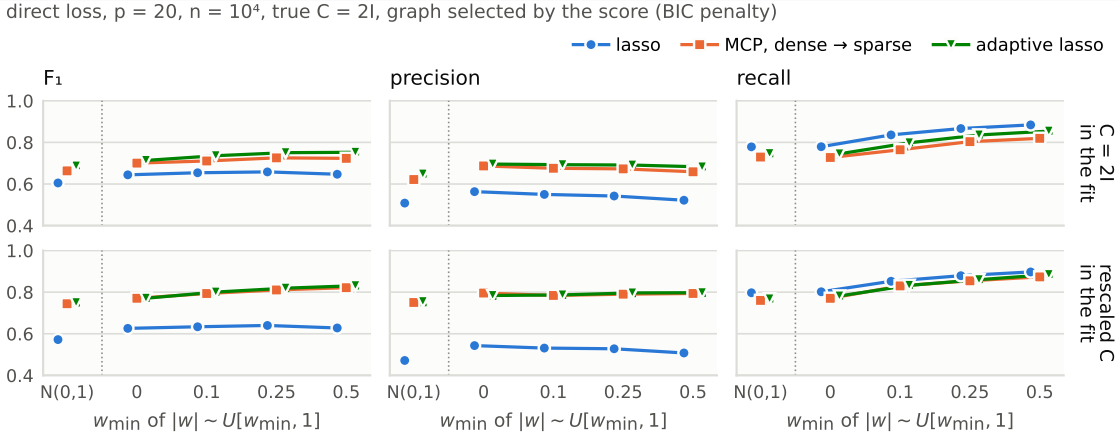
Expected number of edges $(p-1)k$ and diagonal $M_{ii} = -\sum_{j \neq i} |M_{ij}| - |\varepsilon_{ii}|$, $\varepsilon_{ii} \sim N(0, 1)$, as in the thesis generator.

F1, graph class: DAGs against the thesis generator ($n = 10^4$)

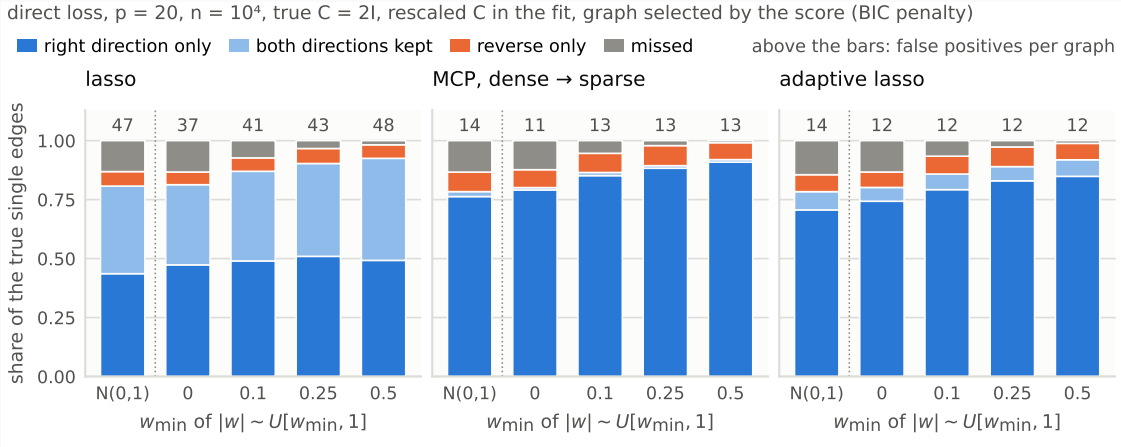


Gain: F_1 minus the lasso's F_1 on the same data, mean over the 25 graphs per k; bars.
With the rescaled C the gain is larger on DAGs at every k; with $C = 2I$ it is about the same.

F2–F3, edge weights: F_1, precision and recall ($n = 10^4$)

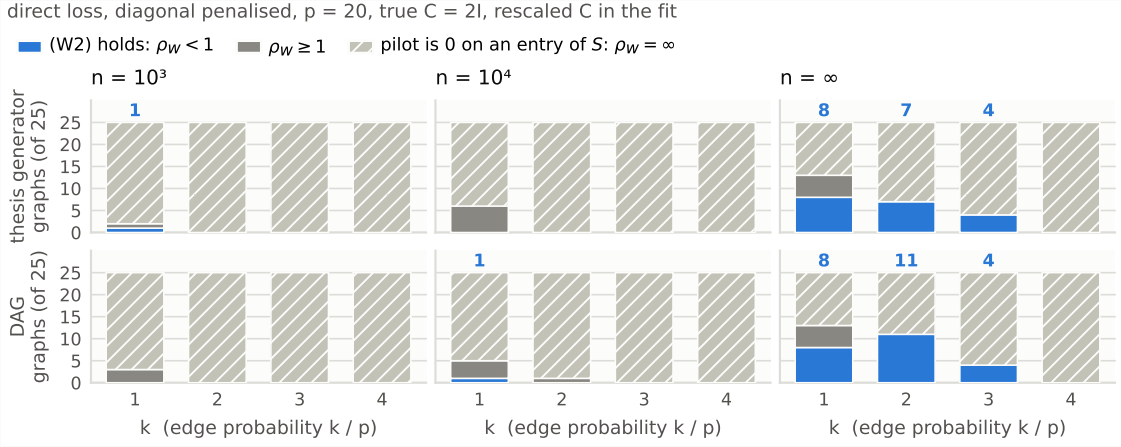


F2–F3, edge weights: what the selected graph keeps of the true edges



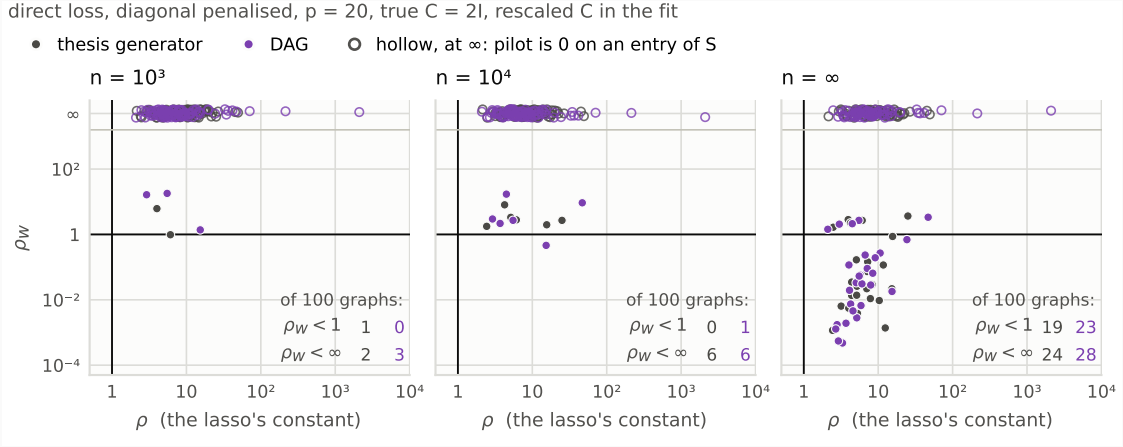
As $w_{\min}$ grows, grey (missed) turns light blue for the lasso, dark blue for the dense-start estimators.

F4–F6, (W2): how often it holds ($p = 20$, rescaled C)



Hatched: the pilot is 0 on an entry of S, so $\bar{w}_S = \infty$ and Theorem W.1 does not apply.
(W2) holds (blue, count above the bar) almost only at $n = \infty$, and never at $k = 4$.

F4–F6, (W2): ρ_w against the lasso's ρ ($p = 20$, rescaled C)



Every graph lies right of $\rho = 1$: the lasso's condition fails in all of them.
Where the pilot keeps every entry of S (filled markers), at $n = \infty$, ρ_w is mostly far below 1.

F7, small DAGs at $n = \infty$: data

Drift: $M^* = -\operatorname{diag}(d_1, \ldots, d_p) + \varepsilon B$, with $B_{ht} = \pm 1$ for each edge $t \to h$ (parent t, child h), else 0

- one weight ε on every edge, $\varepsilon = 0.02, 0.04, \ldots, 6$; the sign is -1 only on $1 \to 2$ in two variants
- $d_i = 1 +$ the length of the longest directed path into i, so $d_t < d_h$ on every edge
- 11 graphs, listed on the next slide

e.g. the chain $1 \to 2 \to 3$: $\quad M^* = \begin{pmatrix} -1 & 0 & 0 \\ \varepsilon & -2 & 0 \\ 0 & \varepsilon & -3 \end{pmatrix}$

Population: true $C = 2I$, $M^*\Sigma^* + \Sigma^* M^{*\top} + C = 0$, $n = \infty$; the diagonal penalised
Fitted to Σ^* with $C = 2I$ (not standardised)

Conditions

- $\rho_{\text{weak}} = \|P \operatorname{sign}(m^*_S)\|_\infty$, which the lasso needs ≤ 1 [1]; ρ_w with the pilot at $\lambda_0 = 10^{-4}\lambda_{\max}$
- $m(0^+)$: the pilot's limit as $\lambda_0 \to 0$, the minimum-ℓ_1 exact fit

F7, small DAGs: the eleven graphs

Names as in the figures; d is the diagonal of $-M^*$, one weight ε on every edge.

graph	edges	d	
chain $1 \to 2 \to 3$	$1 \to 2 \to 3$	$(1, 2, 3)$	
chain of 4	$1 \to 2 \to 3 \to 4$	$(1, 2, 3, 4)$	
chain of 5	$1 \to 2 \to 3 \to 4 \to 5$	$(1, 2, 3, 4, 5)$	
chain, close d	$1 \to 2 \to 3$	$(1, 1.2, 1.44)$	$d_t/d_h \approx 0.83$ on both edges
3 parallel paths	$1 \to \{2, 3, 4\} \to 5$	$(1, 2, 2, 2, 3)$	no edge $1 \to 5$
diamond	$1 \to \{2, 3\} \to 4$	$(1, 2, 2, 3)$	no edge $1 \to 4$
3 paths, $M_{21} < 0$	as 3 parallel paths, $M_{21} = -\varepsilon$	$(1, 2, 2, 2, 3)$	the path via 2 is negative
diamond, $M_{21} < 0$	as diamond, $M_{21} = -\varepsilon$	$(1, 2, 2, 3)$	the path via 2 is negative
collider	$1 \to 3 \leftarrow 2$	$(1, 1, 2)$	two parents, one child
fork	$2 \leftarrow 1 \to 3$	$(1, 2, 2)$	one parent, two children
triangle	$1 \to 2 \to 3$, $1 \to 3$	$(1, 2, 3)$	the chain plus its shortcut $1 \to 3$

F7, small DAGs ($n = \infty$, not standardised): ρ, ρ_{weak} and ρ_w along ε

See Figures "population_constants_standardized.png" and "population_constants_not_standardized.png"

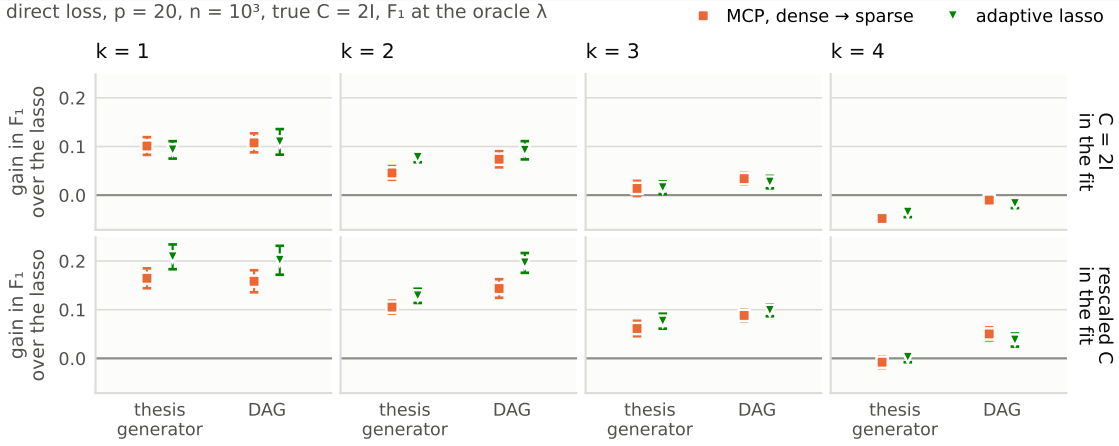
References

[1] Dettling, Drton & Kolar (2024). On the lasso for graphical continuous Lyapunov models. *CLeaR 2024*.

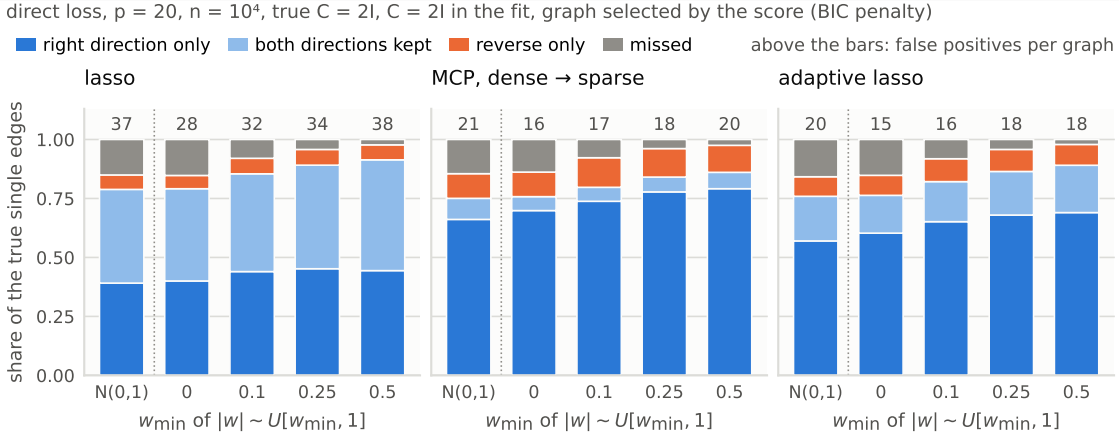
[2] Zou (2006). The adaptive lasso and its oracle properties. *JASA* 101.

[3] Zhou, van de Geer & Bühlmann (2009). Adaptive lasso for high dimensional regression and Gaussian graphical modeling. *arXiv preprint*.

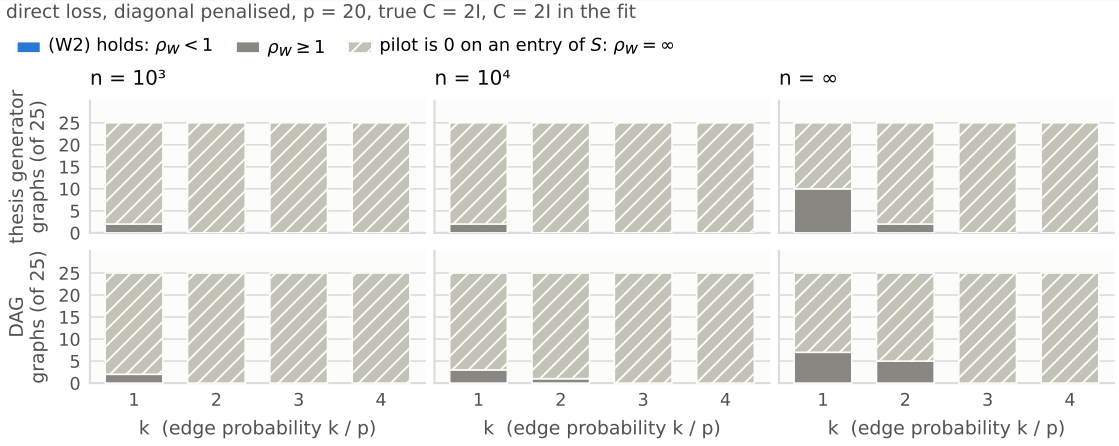
Backup: F1 at $n = 10^3$



Backup: F2–F3 with $C = 2I$ in the fit



Backup: F4–F6 with $C = 2I$ in the fit



Backup: F4–F6, ρ_W against ρ with $C = 2I$ in the fit

